MatLab 1

B13901165張勝勛
班級：王鈺強老師班

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1 Matlab result

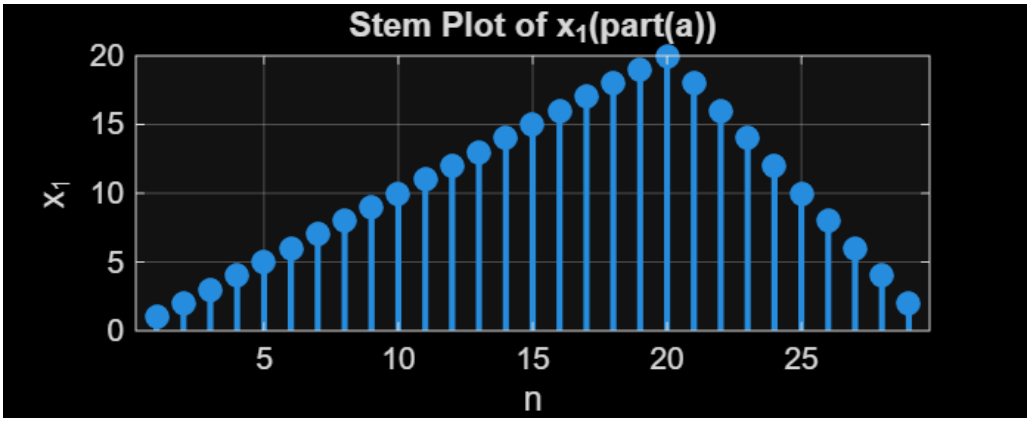

图 1: The result for Part(A)

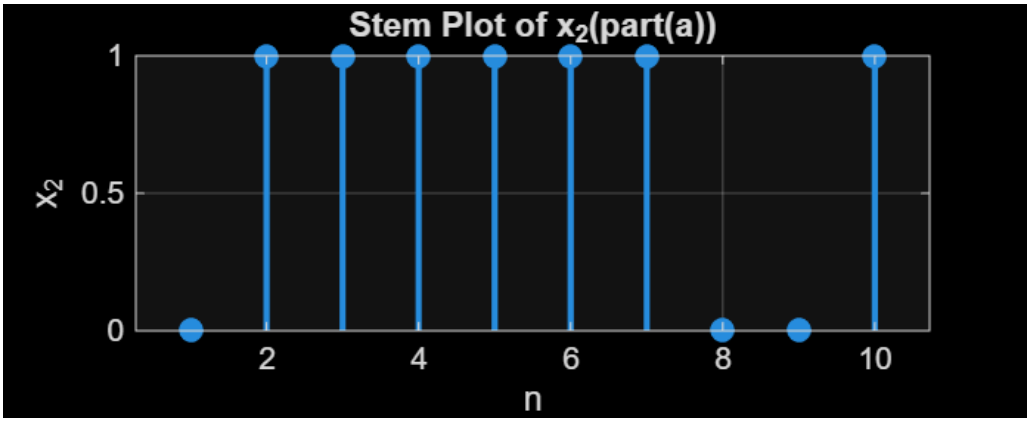

图 2: The result for Part(A)

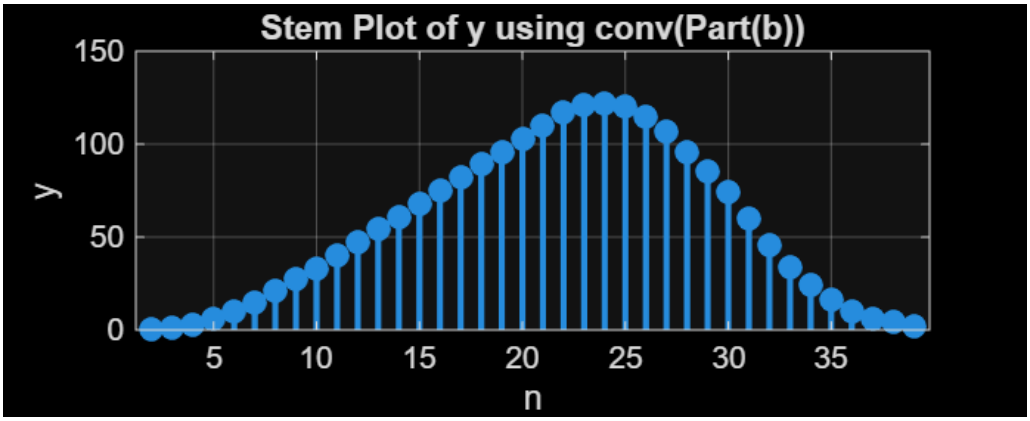

图 3: The result for Part(B)

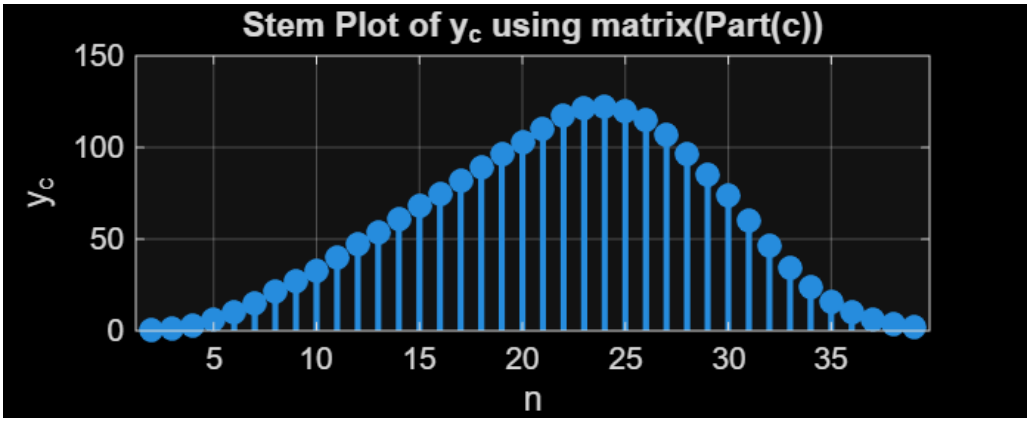

图 4: The result for Part(C)

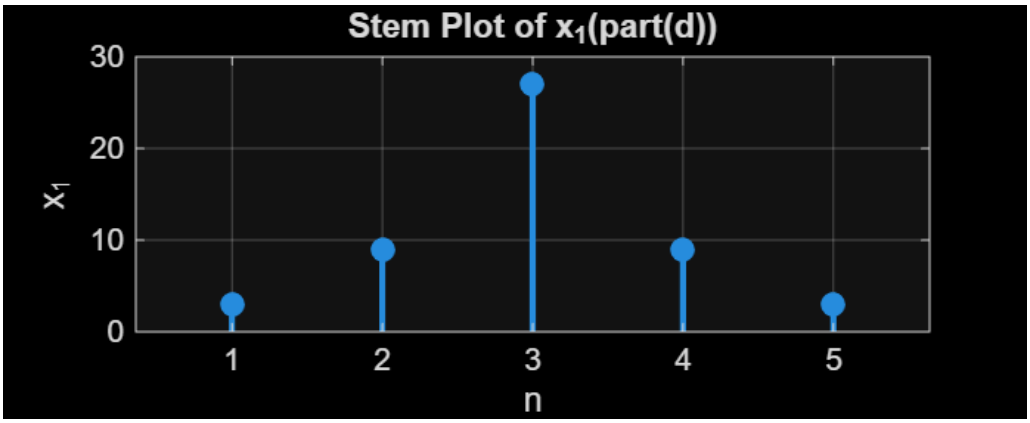

图 5: The result for Part(D)

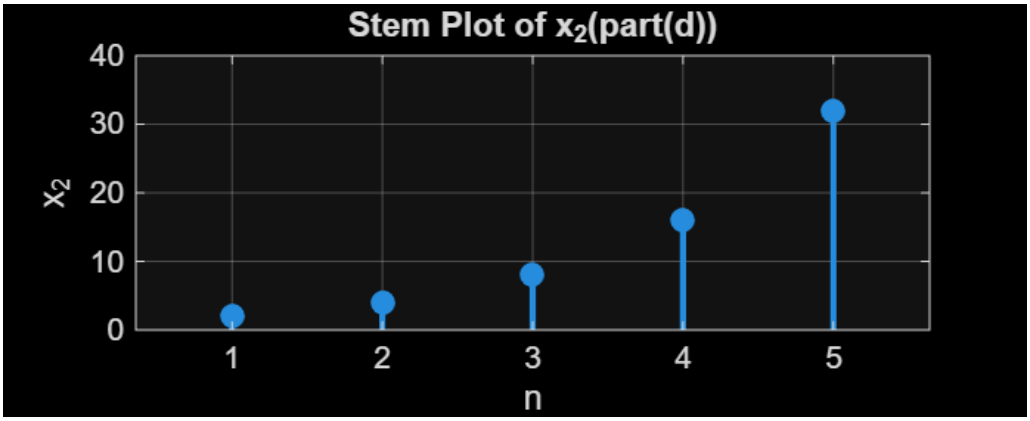

图 6: The result for Part(D)

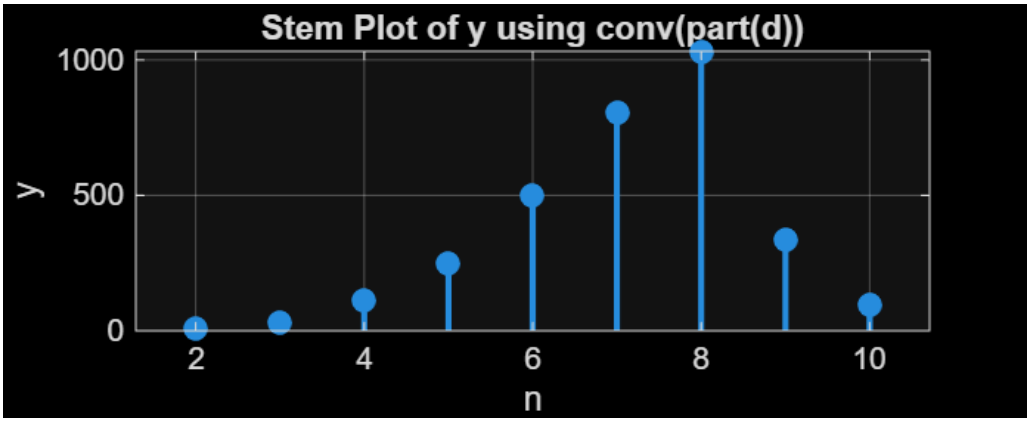

图 7: The result for Part(D)

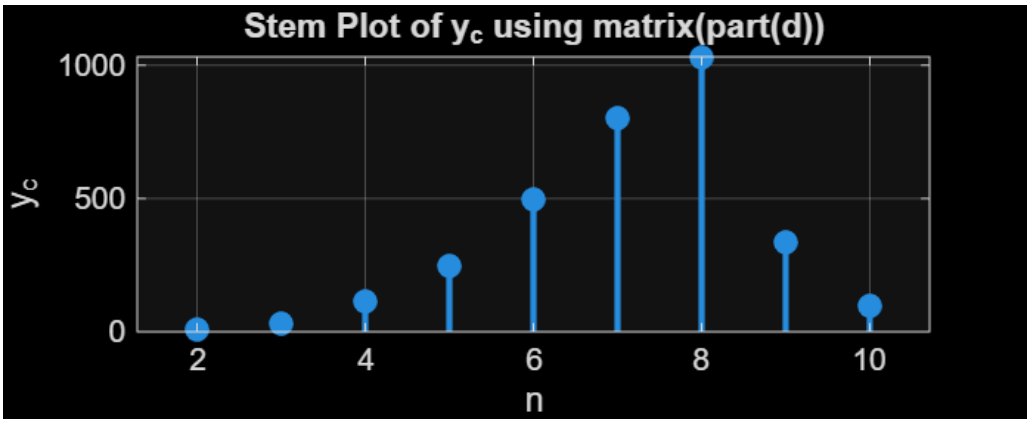

图 8: The result for Part(D)

2 Part(C) code implementation

Listing 1: MATLAB Code

```
%%part(c)
N1 = length(x_1);
N2 = length(x_2);

col = [x_1, zeros(1, N2-1)];
row = [x_1(1), zeros(1, N2-1)];
matrix = toeplitz(col, row);
y_c = matrix * x_2';

if isequal(round(y_c, 5), round(y', 5))
    disp('The result is the same');
end

figure;
stem(n_y, y_c, "filled", "LineWidth", 2);
title('Stem Plot of y_c using matrix(Part(c))');
xlabel('n'); ylabel('y_c');
grid on;
```